Unit 1
1. What is a corpus in NLP, and what are the different types of corpora?
2. Identify and Explain the key components of Natural Language Processing
9. Define TF-IDF and explain how it improves upon Bag of Words
13. Define corpora and discuss their importance in NLP applications.
14. Explain the process and techniques used in word tokenization
21. How does the OneHotEncoder represent textual data numerically?
27. How are regular expressions used in NLP tasks?
33. What is Word2Vec and how does it capture semantic information?
34. Compare and contrast count-based vs. prediction-based text representation methods

Unit 2
3. Describe the steps involved in training a Naive Bayes classifier.
4. How is Naive Bayes optimized specifically for sentiment analysis?
10. Discuss the use of Naive Bayes as a language model.
15. What are precision, recall, and F-measure? Why are they important in evaluating classifiers?
16. What role does smoothing play in Naive Bayes training?
35. Explain how Naive Bayes handles text classification for multiple classes.
36. Compare Naive Bayes with other classification algorithms in NLP.
28. How does the independence assumption impact Naive Bayes classification?

Unit 3
5. What is a Hidden Markov Model (HMM) and how is it applied in speech recognition?
6. Explain the Viterbi algorithm and its role in decoding HMM sequences.
11. What are the steps involved in training a speech recognizer?
17. How is acoustic processing performed on speech signals?
18. Describe advanced methods used for decoding in speech recognition.
30. Discuss the role of attention weights in interpreting Transformer model outputs.
31. Describe techniques used for waveform generation in speech synthesis.
29. How do POS tags contribute to higher-level NLP tasks?
23. What are the steps involved in training a speech recognizer?
24. What are tagsets in POS tagging? Give examples of common tagsets used for English.

Unit 4
7. What is a self-attention network and how does it function within Transformers?
8. Describe the multihead attention mechanism and its advantages.
32. Explain the concept of residual streams in Transformer architectures.
12. How do Transformers overcome limitations of traditional sequential models like RNNs?
19. Discuss how positional encoding is integrated into Transformer models.
20. What are the training objectives commonly used for large language models?